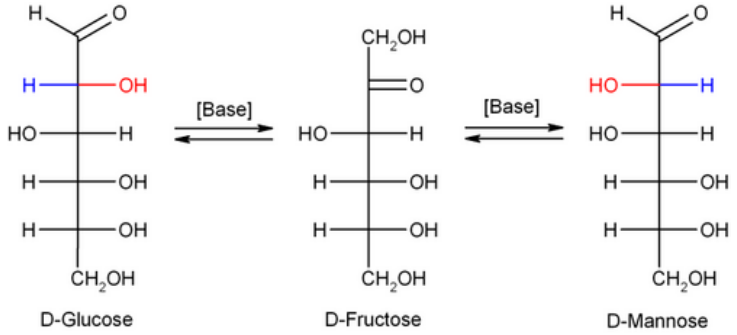
Biochemistry

1. Carbohydrates

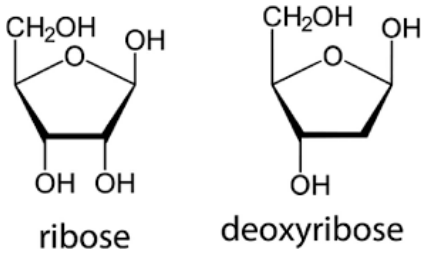
1.1 Monosaccharides

1.1.1 Terminology

Monosaccharides can be classified into several groups based on their structure and length. Those with an aldehyde group are termed aldoses, while those with a ketone group are termed ketoses (the ketone is typically found on the C2 position). Additionally, depending on the number of carbon atoms it contains, it could be classified as a hexose (6C), pentose (5C), and so on. This terminology is explained through the following examples.



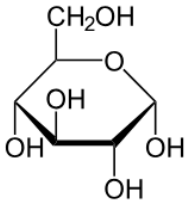
These three sugars are structural isomers as they have the same molecular formula. Glucose and mannose are aldoses, while fructose is a ketose. They can be interconverted through a process known as keto-enol tautomerism, which will be described later on.



Ribose and deoxyribose, found in RNA and DNA, respectively, are pentoses. They have 5 carbon atoms in the carbon chain.

1.1.2 Straight-Chain and Ring Forms

Sugars exist in aqueous solution as a mixture of straight-chain and ring forms (isomers). Glucose is typically drawn as the ring form:



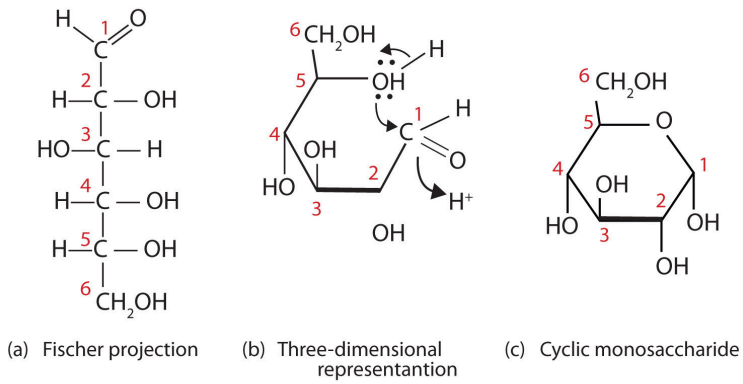
This is accurate - over 99.9% of glucose is in the ring form under normal conditions. The ring form is favoured for two reasons:

1. **Minimises steric hindrance**: the ring form 'locks' substituents (in this case, the hydroxy groups) into fixed positions rather than freely moving around via C-C bond rotations in the straight-chain form. This reduces the steric hindrance between these groups, increasing the stability of the molecule.

2. **Reduces torsional strain**: the ring form allows bond angles to stay close to 109.5°, which is the ideal bond angle for the tetrahedral carbon atoms.

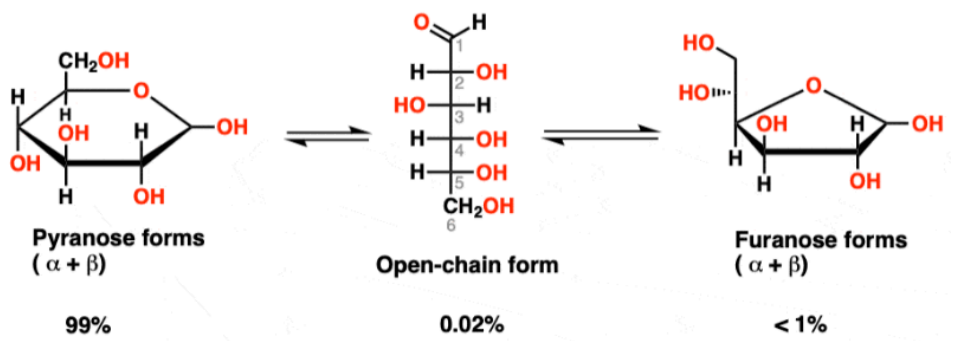
Although the ring form has lower conformational entropy (entropy of spatial shapes) than the straight-chain form, it is more stable due to the enthalpic gain.

The straight-chain and ring forms interconvert rapidly through the nucleophilic addition of the C5 hydroxy group to the C1 aldehyde:



1.1.3 Structural and Optical Isomers of Glucose

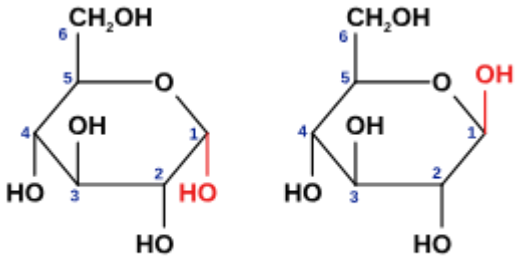
Examining the above image, we notice that the C5 hydroxy group does not need to be the one attacking the aldehyde. It could be, for instance, the C4 hydroxy group, which attacks, forming a five-membered ring instead of the six-membered ring. This is precisely the case:



However, the six-membered ring form (termed 'pyranose') far outweighs the five-membered ring form (termed 'furanose'). This is because the five-membered ring has higher torsional strain since the bond angles created are smaller than the ideal 109.5°.

Notice the presence of the α and β symbols in the above image. Each of these forms (pyranose and furanose) can be further distinguished into two optical isomers. Examine how the ring-closing reaction to convert glucose from the straight-chain to ring form works. The

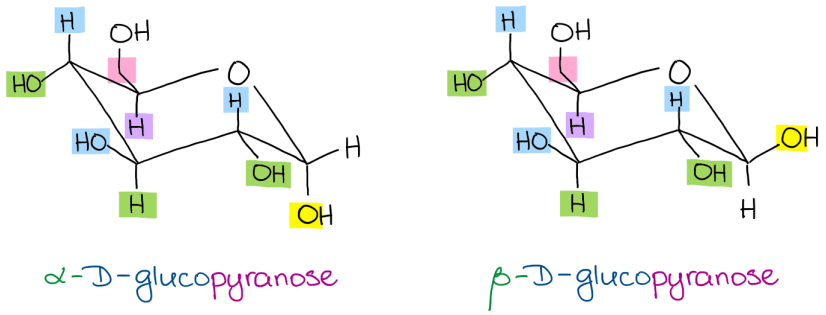
hydroxy group can attack the aldehyde from either the top or bottom face, yielding two different stereoisomers - precisely termed the α and β isomers.



Alpha-(α) Glucose Beta-(β) Glucose

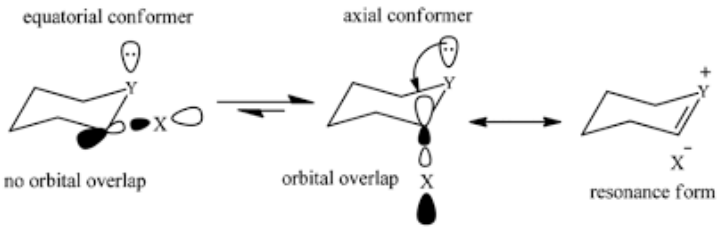
Notice the change in position of the C1 hydroxy group. The C1 carbon atom is called the anomeric carbon, and the hydroxy group attached there is called the anomeric hydroxy group. Isomers which differ only at the anomeric carbon atom (such as the α and β isomers) are termed anomers.

For pyranose (remember, this is glucose in the six-membered cyclic form), the β anomer is preferred over the α anomer in a 64:36 ratio. To understand why, we must look at the chair conformations of each structure (the above images were a simplified diagram taught in many biology textbooks; for a comprehensive understanding of biochemistry, it is highly suggested that you first learn organic chemistry).



The α anomer has the C1 hydroxy group (yellow) in an axial position (vertical), while the β anomer has the hydroxy group in an equatorial position (horizontal). The equatorial position reduces steric hindrance between hydroxy groups on the same molecule (termed 1,3-diaxial interactions) by positioning them facing away from the rest of the molecule. Therefore, the β anomer is more stable than the α counterpart.

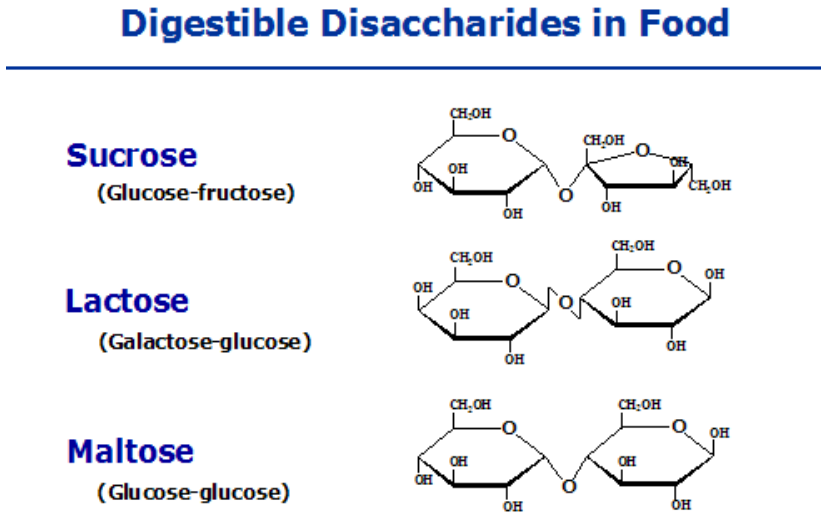
Extra Organic Chemistry Content: The ratio of β:α is predicted to be even greater at 89:11 if not for the anomeric effect. This is an effect whereby a lone pair of electrons on the oxygen atom in the ring can be donated into the σ-antibonding orbital of the C-O bond in the hydroxy group, stabilising it via a hyperconjugative effect. This only occurs in the α anomer where the orbitals align. (You don't have to know all this; this is just for the organic chemistry nerds.)



1.1.4 Keto-Enol Tautomerism

1.2 Disaccharides

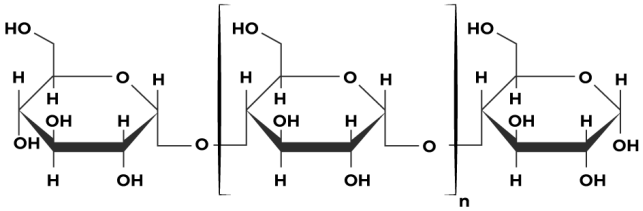
Disaccharides form with a glycosidic bond between the anomeric hydroxy group and another hydroxy group on the other sugar molecule.



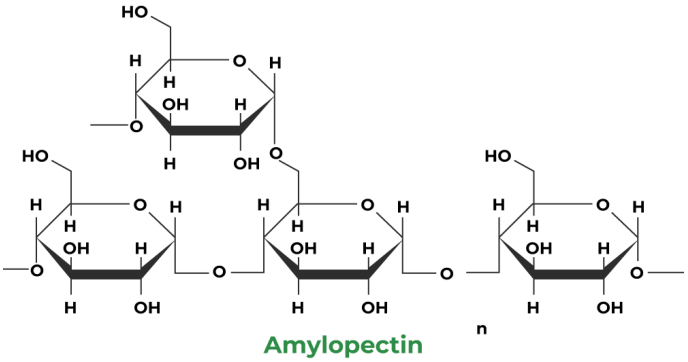
Maltose and lactose have 1,4-glycosidic linkages, meaning the bond is formed between the C1 and C4 hydroxy groups. Sucrose has a 1,2-glycosidic linkage, a special case where the bond is formed between the two anomeric carbon atoms. This means that sucrose is locked in the ring form at neutral or basic pH, as there is no way to convert back to the aldehyde / ketone form (no anomeric hydroxy group to deprotonate in the first step of conversion). It is an acetal group instead of a hemiacetal. So, sucrose is not a reducing sugar, as the reducing properties of a sugar come from the aldehyde group. Therefore, in Benedict's test, sucrose gives a negative result in the basic conditions of the test. However, if sucrose is first acidically hydrolysed before the test, then it will give a positive result since acetals can only be cleaved in acidic conditions.

1.3 Polysaccharides

Starch: Starch contains two chains of sugars, amylose and amylopectin. Amylose comprises around 20-30% of the starch, and consists of long unbranched chains of glucose molecules joined by 1,4-glycosidic linkages. Notably, only alpha glucose is present. Amylopectin comprises the remaining 70-80% and consists of shorter chains of glucose molecules with slight branching. Occasionally, amongst the 1,4-glycosidic linkage chains, there will be a bend which contains a single 1,6-glycosidic linkage, causing the branching.



Amylose



Amylopectin

The functional differences between amylose and amylopectin include: amylopectin is digested faster than amylose, amylopectin is more soluble in water than amylose, amylopectin is sweeter and has a higher glycaemic index.

Starch forms a thick translucent gel when submerged in water with heating. This occurs in several steps. Firstly, water is absorbed into the amorphous space in the starch, causing it to swell. Then, it travels from this space to the amylopectin, causing the amylopectin to dissolve by forming hydrogen bonds with it. This allows the structure of the starch particle to become less compact, allowing some amylose molecules to leach into the surrounding water and increase the viscosity, forming the gel-like substance.

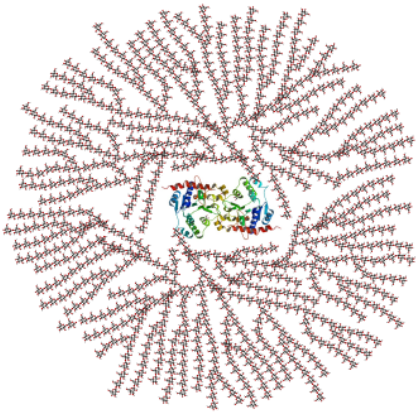
Starch is used as a food storage unit in plants (in the form of granules) as it is more compact and also does not exert any osmotic pressure since it is insoluble, allowing high quantities to accumulate.

Glycogen: glycogen is the storage unit of glucose in animal cells. The structure of glycogen is similar to amylopectin, except it is much more branched. Glycogen exists in balls with a core of a glycogenin protein, which acts as a priming protein to kickstart the formation of glycogen globules. The steps are as follows:

1. Glycogenin binds glucose from UDP-glucose (uridine diphosphate glucose) up to eight times in total for each glycogenin molecule.

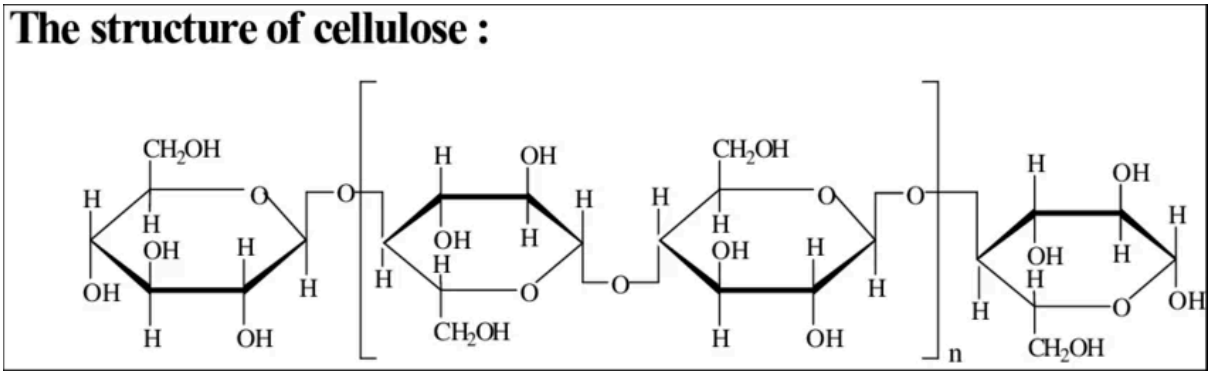
2. Glycogen synthase takes over, adding more glucose molecules to this initial ball (glycogen synthase can only add to an existing chain of 3 or more glucose molecules, thus glycogenin is required prior)

3. The addition continues until a globule of around 30000 glucose units is formed.



Animals use glycogen instead of starch as they require more energy in a shorter amount of time. Glycogen is more compact for higher energy density, and also it has more branching, which provides more free ends for hydrolysis to release glucose at a higher rate. Glycogen stores in humans could be depleted in a day unless replenished through eating, which is why other stores of food, such as fats, are required as well. Similar to starch, glycogen only consists of the alpha anomer of glucose.

Cellulose: Cellulose is a structural component of plants. It consists of long straight chains of beta anomer of glucose molecules bonded together by 1,4-glycosidic linkages. However, since in the beta anomer, the anomeric hydroxy group is oriented in the equatorial position while the C4 hydroxy group is in the axial position, to maintain a straight chain, alternate beta-glucose molecules must be upside down with respect to their neighbours. If not, then the chains would be aligned in a diagonal manner.



Cellulose strands are bonded to adjacent strands by hydrogen bonds, and the bundle of about 80 cellulose strands is called a microfibril. Microfibrils are present in cell walls and provide mechanical support for plants.

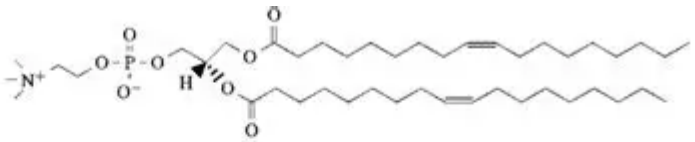
Enzymes which digest starch are unable to digest cellulose due to the differing linkages (alpha vs beta). In humans, cellulose is the insoluble fibre on nutrition labels which adds bulk to food, thus increasing abrasion to the walls of the digestive tract and stimulating the secretion of mucus, which helps food pass smoothly through the tract.

Chitin: Chitin is the component of arthropod exoskeletons (insects, spiders, crustaceans, etc.). It has a similar structure to cellulose with the beta linkages, except that each glucose monomer of chitin has an acetamido ($-NHCOCH_3$) group instead of the hydroxy group on C2. Hydrolysis of chitin to remove the acetyl group and leave the amine gives chitosan, a polymer used in the medical sector for the production of antibacterial bandages.

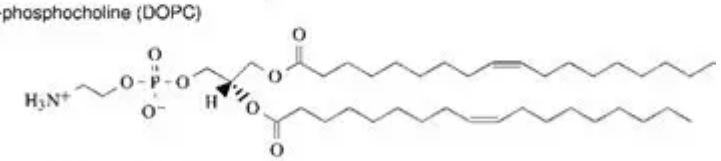
2. Lipids

Lipids or fats are a class of macromolecules which share the property of hydrophobicity. There are many types of lipids (e.g., triglycerides, phospholipids, steroids).

Phospholipids: One of the fatty acid chains on glycerol is replaced with a phosphate group (forming a phosphate ester). Often, there is an additional attachment, such as choline, to the phosphate group, forming a phosphodiester bond. The phosphate group is negatively charged and thus is hydrophilic, in contrast to the hydrophobic fatty acid chains.



(a) 1,2-dioleoyl-sn-glycero-3-phosphocholine (DOPC)



(b) 1,2-dioleoyl-sn-glycero-3-phosphoethanolamine (DOPE)

Therefore, phospholipids arrange themselves in a bilayer when forming membranes, with the fatty acid tails pointing inward.

Fatty Acid Synthesis (de novo lipogenesis): Conversion of glucose into fatty acids for storage as fats is detailed below. The first step is the carboxylation of acetyl-CoA by acetyl-CoA carboxylase to form malonyl-CoA. The acetyl-CoA from the link reaction (oxidative decarboxylation) cannot be directly transported from the mitochondria into the cytoplasm, where the synthesis of fatty acids and cholesterol takes place. Instead, citrate is transported to the cytosol, where it is cleaved by ATP citrate lyase into acetyl-CoA and oxaloacetate (essentially the reverse of the step in the Krebs cycle). Finally, fatty acid synthase extends the carbon chain to form the fatty acid.

3. Proteins

Proteins must be distinguished from polypeptides; the former refers to the specific arrangement and conformation of one or more polypeptides to serve a specific purpose, while the polypeptide itself simply refers to the specific sequence of amino acids.

There are 4 structural levels of proteins:

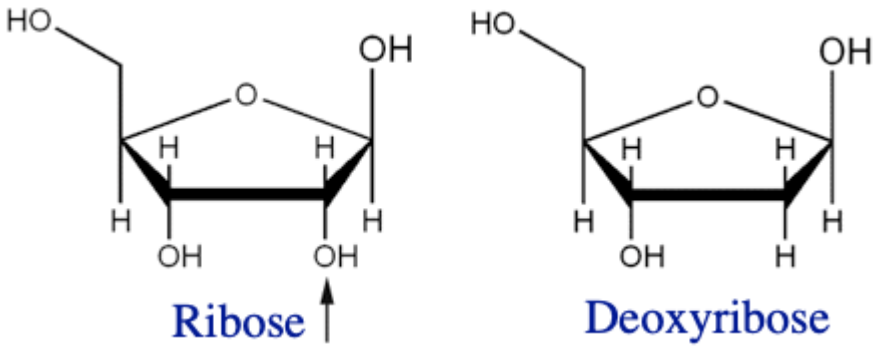
- Primary: Linear chain of amino acids, this is just the specific sequence of amino acids used to form a polypeptide chain.
- Secondary: Regions stabilised by hydrogen bonds between atoms on the polypeptide backbone. This is a 3D arrangement of the polypeptide chain based on intramolecular forces of attraction between groups on the polypeptide backbone (note that this only consists of the peptide linkages and not the side chain interactions). The hydrogen bonds form between the hydrogen and oxygen atoms of the amides.
    - Alpha helix: Coil held together by hydrogen bonds between every fourth amino acid

    - Beta pleated sheet: two or more segments of the polypeptide that lie side by side

- Tertiary: 3-dimensional shape stabilised by interactions between side chains
    - Hydrophobic interaction: Non-polar side chains converge in the middle of the 3D structure, while polar side chains face outward due to the differing levels of attraction to the external water molecules. Van der Waals forces hold the non-polar side chains in the middle together.
    - Disulfide bridges: when two cysteine molecules bond together using their thiol groups, releasing hydrogen atoms and forming the R-S-S-R linkage (stronger bond as it is covalent)
- Quaternary: Association of two or more polypeptides (only applies to proteins with more than one polypeptide unit)
    - Collagen consists of 3 identical helical polypeptides intertwined into a larger triple helix, giving the fibres high strength.
    - Haemoglobin consists of 2 alpha and 2 beta polypeptide subunits. Each subunit has a non-polypeptide component called heme, which contains an iron atom that O_2 molecules coordinate to.

4. Nucleotides

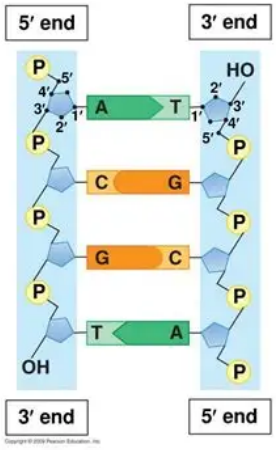
Nucleic acids are macromolecules consisting of nucleotides. A nucleotide consists of 1 to 3 phosphate groups bonded to a nucleoside, which in turn consists of a 5-carbon sugar and a nitrogenous base. The 5-carbon sugar is ribose in RNA and deoxyribose in DNA, differing only in the presence of an oxygen atom at the C2 position.



In DNA and RNA, there is only 1 phosphate group bonded to the C5 hydroxy group, whereas in other nucleotides, such as ATP and ADP, there can be 3 or 2 respectively.

A nitrogenous base, either a purine (adenine and guanine) or a pyrimidine (thymine, uracil or cytosine), is attached to the hydroxy group on the C1 position of the sugar. Thymine is used in DNA, while uracil is used in RNA.

There are 2 ends of a strand of DNA or RNA, the 5’ and 3’ ends. The 5’ end refers to the C5 position bearing the final phosphate group, while the 3’ end refers to the C3 position with the unmodified hydroxy group remaining there.



DNA is arranged in a double helix between the two antiparallel strands. RNA can also form 3D structures; for instance, tRNA takes on an L-shaped structure.

Metabolism is split into two categories: catabolism, which is the degradation of complex molecules into simpler molecules while releasing energy, and anabolism, which is the reverse and consumes energy.

4.2 ATP

ATP is hydrolysed to ADP to release energy (the energy released in the formation of the O-H and P-OH bonds in the HPO_4^{2-} and ADP, respectively, is greater than the energy required to split the O-H bond in water). ATP is useful as it releases a greater amount of energy than the hydrolysis of the same bonds in another molecule normally would. This is due to the decrease in repulsion between the negative charges on adjacent phosphate groups in the ATP molecule when it is hydrolysed.

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